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Toner supply device and image forming apparatus

Abstract

A toner supply device includes toner containers each containing toner used to form a latent image on a latent-image bearer, and circuitry to perform toner supply control to supply toner from the toner containers to developing devices. In the toner supply device, the toner supply control includes repeating toner supply in which one of the toner containers supplying the toner is switched to another one of the toner containers when a supplying time to supply the toner from one of the toner containers to corresponding one of the developing devices exceeds a prescribed maximum supplying time. In the toner supply device, the prescribed maximum supplying time, in the toner supply performed for a first time after the toner supply control starts, is shorter than the prescribed maximum supplying time in the toner supply performed for a second time.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application Nos. 2023-045645 and 2023-184210, filed on Mar. 22, 2023, and Oct. 26, 2023, respectively, in the Japan Patent Office, the entire disclosures of which are hereby incorporated by reference herein.

BACKGROUND

Technical Field

(2) Embodiments of the present disclosure relate to a toner supply device and an image forming apparatus.

Background Art

(3) In the related art, toner supply devices have been proposed each of which is provided with a plurality of toner containers that contain toner used to form a latent image on a latent-image bearer. Such toner supply devices perform toner supply control in which toner is supplied from a plurality of toner containers to corresponding developing devices.

(4) Moreover, such toner supply devices sequentially switch a plurality of toner containers that

supply toner and supply toner step by step to the developing device that needs toner supply. Due to such a configuration, toner is not supplied to a plurality of toner containers at the same time.

SUMMARY

(5) Embodiments of the present disclosure described herein provide a toner supply device including toner containers each containing toner used to form a latent image on a latent-image bearer, and circuitry to perform toner supply control to supply toner from the toner containers to developing devices. In the toner supply device, the toner supply control includes repeating toner supply in which one of the toner containers supplying the toner is switched to another one of the toner containers when a supplying time to supply the toner from one of the toner containers to corresponding one of the developing devices exceeds a prescribed maximum supplying time. In the toner supply device, the prescribed maximum supplying time, in the toner supply performed for a first time after the toner supply control starts, is shorter than the prescribed maximum supplying time in the toner supply performed for a second time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more complete appreciation of embodiments and the many attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings.
- (2) FIG. 1 is a schematic diagram of a copier according to an embodiment of the present disclosure.
- (3) FIG. 2 is a diagram illustrating a schematic configuration of an image forming unit that handles a yellow toner image, according to an embodiment of the present disclosure.
- (4) FIG. 3 is a schematic sectional view of a color-printing driver device according to an embodiment of the present disclosure.
- (5) FIG. 4 is a schematic diagram of a C color toner supply unit of a cyan (C) and black (K) color toner supply device, according to an embodiment of the present disclosure.
- (6) FIG. 5 is a schematic partial sectional view of a C color toner supply unit according to an embodiment of the present disclosure.
- (7) FIG. 6 is a schematic diagram illustrating how driving force is conveyed to a K color toner supply unit, according to an embodiment of the present disclosure.
- (8) FIG. 7 is a schematic diagram illustrating how the driving force is conveyed to a C color toner supply unit, according to an embodiment of the present disclosure.
- (9) FIG. 8 is a diagram illustrating the hardware blocks of a copier according to an embodiment of the present disclosure.
- (10) FIG. 9 is a flowchart of the processes of setting the drive time of a toner supply unit of each color, according to an embodiment of the present disclosure.
- (11) FIG. 10A is a timing chart of toner supply control in which a desired amount of toner is supplied to each developing device in one-time toner supply, according to an embodiment of the present disclosure.
- (12) FIG. 10B is a timing chart of toner supply control in which a desired amount of toner is supplied to each developing device in the toner supply performed multiple times, according to an embodiment of the present disclosure.
- (13) FIG. 11 is a timing chart to explain a problem of toner supply control when the shortest image-forming operation is performed continuously a plurality of times, according to the related art.
- (14) FIG. 12 is a timing chart of toner supply control according to an embodiment of the present disclosure.
- (15) FIG. 13 is a flowchart of toner supply control according to an embodiment of the present disclosure.

(16) FIG. 14 is a flowchart of toner supply according to an embodiment of the present disclosure.
(17) The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted.

DETAILED DESCRIPTION

(18) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to limit the present disclosure. As used herein, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes” and/or “including”, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(19) In describing example embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the present disclosure is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have the same structure, operate in a similar manner, and achieve a similar result.

(20) Some embodiments of the present disclosure are described below with reference to the drawings. Note that numerous additional modifications and variations are possible in light of the above teachings. It is therefore to be understood that within the scope of the appended claims, the embodiments of the present disclosure may be practiced otherwise than as specifically described herein. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of this disclosure and appended claims. The above-described embodiments are illustrative and do not limit the present disclosure.

(21) A copier **500** according to an embodiment of the present disclosure that serves as an image forming apparatus is described below.

(22) FIG. 1 is a schematic diagram of a copier **500** according to an embodiment of the present disclosure.

(23) Typically, the copier **500** includes the printer unit **100**, a sheet feeder **200**, and an image reading unit **400** arranged above the printer unit **100**. In the following description, the sheet feeder **200** may be referred to as a sheet feeding table.

(24) A toner container mount **70** is arranged at an upper portion of the printer unit **100** to hold four detachable and replaceable toner containers **32Y**, **32M**, **32C**, and **32K** that serve as four powder containers and handle multicolor toner including yellow (Y), magenta (M), cyan (C), and black (BK). An intermediate transfer unit **85** is arranged below the toner container mount **70**.

(25) The intermediate transfer unit **85** according to the present embodiment includes, for example, an intermediate transfer belt **48**, four primary-transfer bias rollers **49Y**, **49M**, **49C**, and **49K**, a secondary-transfer backup roller **82**, a plurality of tension rollers, and an intermediate-transfer cleaning device. The intermediate transfer belt **48** is stretched and supported by the above-described multiple rollers, and is seamlessly moved in the direction indicated by an arrow in FIG. 1 as the secondary-transfer backup roller **82**, which is one of the multiple rollers, is driven to rotate.

(26) In the printer unit **100**, four image forming units **46Y**, **46M**, **46C**, and **46K** that handle toner images of different colors including yellow, magenta, cyan, and black colors are arranged in parallel, so as to face the intermediate transfer belt **48**.

(27) The printer unit **100** further includes an exposure device **47** as a latent image forming unit below the four image forming units **46**. The exposure device **47** according to the present embodiment exposes the surface of a photoconductor **41** as will be described later in detail to form an electrostatic latent image on the surface of the photoconductor **41**, based on the image data of document images read by the scanner unit **400** or the image data that is input from an external

device such as a personal computer (PC). Although the exposure device **47** provided for the printer unit **100** according to the present embodiment employs laser beam scanning using a laser diode, a different method that employs, for example, a light-emitting diode (LED) array may be adopted.

(28) FIG. **2** is a diagram illustrating a schematic configuration of the image forming unit **46Y** that handles the yellow toner image, according to the present embodiment.

(29) The image forming unit **46Y** includes a photoconductor **41Y** that is shaped like a drum and serves as a latent image bearer. In the image forming unit **46Y**, for example, a charging roller **44Y** that serves as a charging device, a developing device **50Y**, a photoconductor cleaning device **42Y** to clean the photoconductor **41Y**, and an electric-charge removing device are arranged around the photoconductor **41Y**. A series of image forming processes including a charging process, an exposure process, a developing process, a transfer process, and a cleaning process are performed on the photoconductor **41Y**. As a result, a yellow toner image is formed on the surface of the photoconductor **41Y**.

(30) As illustrated in FIG. **2**, the developing device **50Y** according to the present embodiment includes, for example, a developing roller **51Y**, a doctor blade **52Y**, and a pair of developer conveying screws **55Y**. The developing roller **51Y** faces the photoconductor **41Y**, and the doctor blade **52Y** faces the developing roller **51Y**. The two developer conveying screws **55Y** are arranged inside two developer storages including a first developer storage **53Y** and a second developer storage **54Y**. The developing roller **51Y** includes, for example, a stationary inner magnet roller (or multiple magnets) and a sleeve that rotates around the magnet roller. The first developer storage **53Y** and the second developer storage **54Y** store developer G containing two components including carriers and toner. The second developer storage **54Y** communicates, through an opening on the upper side thereof, with a downward toner conveyance path **64Y**.

(31) Inside the developing device **50Y**, the developer G is stirred by the two developer conveying screws **55Y** and circulated between the first developer storage **53Y** and the second developer storage **54Y**. While being conveyed by one of the pair of developer conveying screws **55Y**, the developer G in the first developer storage **53Y** is attracted by the magnetic fields generated or formed by the magnet roller inside the developing roller **51Y**, and is fed and borne onto the sleeve surface of the developing roller **51Y**. The sleeve of the developing roller **51Y** is driven to rotate in the counterclockwise direction as indicated by an arrow in FIG. **2**, and the developer G that is borne on the developing roller **51Y** moves on the developing roller **51Y** with the rotation of the sleeve. Under such conditions, the toner particles in the developer G are charged through friction with carrier particles in the developer G to have a potential in the reversed polarity of the polarity of the carrier particles. Then, the toner particles are electrostatically attracted to the carrier particles, and are borne on the developing roller **51Y** together with the carrier particles attracted by the magnetic field generated or formed on the developing roller **51Y**.

(32) The developer G that is borne on the developing roller **51Y** is conveyed in a direction indicated by an arrow in FIG. **2** to a gap between the doctor blade **52Y** and the developing roller **51Y**. When the developer G on the developing roller **51Y** passes through the portion, the amount of the developer G is optimized, and then the developer G is conveyed to a developing region which is a position facing the photoconductor **41Y**. In the developing area, the toner in the developer G adheres to the latent image formed on the photoconductor **41Y** due to the effect of the developing electric field generated between the developing roller **51Y** and the photoconductor **41Y**. As the sleeve rotates, the developer G that has passed through the developing region and remains on the surface of the developing roller **51Y** reaches an area above the first developer storage **53Y** and then drops from the developing roller **51Y**.

(33) The concentration of toner in the developer G that is contained in the developing device **50Y** is adjusted within a predetermined range. More specifically, a toner supply device as will be described later in detail supplies the toner from the toner container **32Y** to the second developer storage **54Y** according to the consumption of the toner included in the developer G in the

developing device **50Y** The toner that is supplied to the second developer storage **54Y** is circulated by the pair of developer conveying screws **55Y** between the first developer storage **53Y** and the second developer storage **54Y** while being mixed with the developer G and stirred.

(34) The other three image forming units **46M**, **46C**, and **46K** have a similar configuration to that of the yellow image forming unit **46Y** except for the color of the toner used therein and form toner images of the respective colors on the photoconductors **41M**, **41C**, and **41K**. In the following description, only the image forming unit **46Y** that handles yellow toner is described, and the description of the other three image forming units **46M**, **46C**, and **46K** are omitted where appropriate.

(35) As illustrated in FIG. 2, the photoconductor **41Y** is driven to rotate in a clockwise direction. The surface of the photoconductor **41Y** is then evenly charged by the charging roller **44Y** at a position facing the charging roller **44Y**. Such a process is referred to as a charging process. When the photoconductor **41Y** reaches a position to receive the laser beam L emitted from an exposure device **47**, the photoconductor **41Y** is scanned with the laser beam L, and thus an electrostatic latent image for yellow is formed thereon. Such a process is referred to as an exposure process. Then, the photoconductor **41Y** reaches a position facing the developing device **50Y**, where the electrostatic latent image is developed with yellow toner into a yellow toner image. Such a process is referred to as a developing process.

(36) The four primary-transfer bias rollers **49Y**, **49M**, **49C**, and **49K** of the intermediate transfer unit **85** sandwich the intermediate transfer belt **48** with the corresponding photoconductors **41Y**, **41M**, **41C**, and **41K**, respectively, to form primary transfer nips therebetween. To the primary-transfer bias rollers **49Y**, **49M**, **49C**, and **49K**, a primary transfer bias whose polarity is opposite to the polarity of the electric charge of the toner is applied.

(37) When the surface of the photoconductor **41Y** on which the toner image has been formed in the developing process reaches the position facing the primary-transfer bias roller **49Y** across the intermediate transfer belt **48**, the toner image is transferred from the photoconductor **41Y** onto the intermediate transfer belt **48** in the primary transfer nip. Such a transferring process is referred to as a primary transferring process. In this step, a small amount of toner remains untransferred on the photoconductor **41Y**. After the toner image is transferred onto the intermediate transfer belt **48** at the primary transfer nip, the surface of the photoconductor **41Y** reaches a position facing the photoconductor cleaning device **42Y**. The untransferred toner that remains on the photoconductor **41Y** at the facing position is mechanically collected by the cleaning blade **42a**. Such a process may be referred to as a cleaning step in the following description. Finally, the surface of the photoconductor **41Y** reaches a position facing the electric-charge removing device, and the residual potential on the photoconductor **41Y** is removed at that position. In this way, a series of image forming processes performed on the photoconductor **41Y** is completed.

(38) Such a series of image forming processes is also performed in the other image forming units **46M**, **46C**, and **46K** in a similar manner to the image forming unit **46Y** that handles yellow toner. In other words, the exposure device **47** disposed below the image forming units **46M**, **46C**, and **46K** irradiates photoconductors **41M**, **41C**, and **41K** of the image forming units **46M**, **46C**, and **46K** with laser beams L based on image data. More specifically, the exposure device **47** includes light sources to emit the laser beams L, a plurality of optical elements, and a polygon mirror that is rotated by a motor. The exposure device **47** uses the optical elements to irradiate the photoconductors **41M**, **41C**, and **41K** with the laser beams L while deflecting the laser beams L with a polygon mirror. Then, after the developing process, the multicolor toner images are transferred from the photoconductors **41M**, **41C**, and **41K** onto the intermediate transfer belt **48** and superimposed on top of one another.

(39) While rotating and moving in the direction indicated by the arrow illustrated in FIG. 1, the intermediate transfer belt **48** sequentially passes through the primary transfer nips of the primary-transfer bias rollers **49Y**, **49M**, **49C**, and **49K**. Thus, multicolor toner images including yellow,

magenta, cyan, and black toner images are primarily transferred from the photoconductors **41Y**, **41M**, **41C**, and **41K** to the intermediate transfer belt **48** on top of one another to form a color toner image on the intermediate transfer belt **48**.

(40) The intermediate transfer belt **48**, to which a color toner image has been formed upon transferring multicolor toner images thereto and superimposing those toner images on top of one another, reaches a position facing a secondary transfer roller **89** arranged opposite the secondary-transfer backup roller **82**. At this position, the secondary-transfer backup roller **82** sandwiches the intermediate transfer belt **48** with the secondary transfer roller **89** to form a secondary transfer nip. The color toner image that is borne on the intermediate transfer belt **48** is transferred onto the sheet P such as a transfer sheet conveyed to the position of the secondary transfer nip. In such a transferring process, the untransferred toner that failed to be transferred to the sheet P remains on the intermediate transfer belt **48**. The intermediate transfer belt **48** having passed through the secondary transfer nip reaches the position of an intermediate transfer cleaning device, and untransferred toner on the surface of the intermediate transfer belt **48** is collected. Thus, a series of transfer processes that are performed on the intermediate transfer belt **48** ends.

(41) The movement of the sheet P is described below. The sheet P is conveyed from a sheet tray **26** of a sheet feeder **200** arranged below the printer unit **100** to the above-described secondary transfer nip through, for example, a sheet feed roller **27** and a registration roller pair **28**. More specifically, a plurality of sheets P are stacked and stored in the sheet tray **26**. When the sheet feed roller **27** is driven to rotate counterclockwise in FIG. **1**, the uppermost sheet P is conveyed toward a roller nip formed by two rollers of the registration roller pair **28**.

(42) The sheet P that is conveyed to the registration roller pair **28** stops moving at the roller nip of the registration roller pair **28** that stops rotating temporarily. The registration roller pair **28** is driven to rotate in accordance with the timing at which the color toner image on the intermediate transfer belt **48** reaches the secondary transfer nip. Accordingly, the sheet P is conveyed toward the secondary transfer nip. Thus, a multicolor toner image is transferred onto the sheet P as desired.

(43) The sheet P onto which the color toner image has been transferred at the secondary transfer nip is conveyed to the position of a fixing device **86**. In the fixing device **86**, the color toner image transferred to the surface of the sheet P is fixed onto the recording medium P by heat and pressure from a fixing belt and a pressure roller. The sheet P having passed through the fixing device **86** is ejected to the outside of the copier **500** through the rollers of an output roller pair **29**. The sheet P ejected to the outside of the copier **500** by the output roller pair **29** is sequentially stacked on the stacking unit **30** as an output image. Thus, a series of image forming processes performed in the copier **500** is completed.

(44) The printer unit **100** according to the present embodiment is provided with a pair of toner supply devices. One of the pair of toner supply devices is a yellow (Y) and magenta (M) color toner supply device **60YM** that supplies toner to the developing device **50Y** of the image forming unit **46Y** and the developing device **50M** of the image forming unit **46M**. The other one of the pair of toner supply devices is a cyan (C) and black (K) color toner supply device **60CK** that supplies toner to the developing device **50C** of the image forming unit **46C** and the developing device **50K** of the image forming unit **46K**.

(45) The YM color toner supply device **60YM** includes a yellow (Y) color toner supply unit **160Y** that supplies the toner of yellow (Y) color stored in the yellow (Y) color toner container **32Y**, which serves as a toner container containing yellow (Y) color toner, to the developing device **50Y** that handles yellow (Y) color toner. The YM color toner supply device **60YM** includes a magenta (M) color toner supply unit **160M** that supplies the magenta (M) color toner stored in the magenta (M) color toner container **32M**, which serves as a toner container containing magenta (M) color toner, to the developing device **50M** that handles magenta (M) color toner.

(46) The CK color toner supply device **60CK** is provided with a C color toner supply unit **160C** that supplies the toner of cyan (C) color to the developing device **50C** that handles the cyan (C)

color toner stored in the cyan (C) color toner container **32C** that serves as a toner container that contains cyan (C) color toner container. The CK color toner supply device **60CK** is provided with a K color toner supply unit **160K** that supplies the toner of black (K) color to the developing device **50K** that handles the black (K) color toner stored in the black (K) color toner container **32K** that serves as a toner container that contains black (K) color toner. The toner supply devices will be described later in detail.

(47) The printer unit **100** according to the present embodiment is provided with a toner adhesion-amount sensor **310** that serves as a density sensor and detects the density of the toner image formed on the outer circumferential surface of the intermediate transfer belt **48**. For example, for-detection image patterns are formed on the surface of the intermediate transfer belt **48** at prescribed timings during continuous printing or every time a predetermined number of sheets are printed, and the amount of adhered toner in the image patterns is detected by the toner adhesion-amount sensor **310**. Subsequently, the image-forming condition such as a developing bias is individually adjusted based on the results of detection for each one of the image forming units, or forced toner supply control as will be described later in detail is performed.

(48) FIG. **3** is a schematic sectional view of a color-printing driver device that drives color photoconductors **41C**, **41M**, and **41Y** and the rotors of the color developing devices **50Y**, **50M**, and **50C** including the developing roller **51** and the developer conveying screw **55**, according to the present embodiment.

(49) The transmission mechanisms of the driving force of the multiple image forming units are substantially equivalent to each other. In view of these circumstances, only the transmission mechanism of the driving force of the image forming unit **46C** that handles cyan (C) color toner is described.

(50) The color-printing driver device **110** according to the present embodiment is provided with a color image-forming motor **1**, and the color image-forming motor **1** is fixed to the surface of the bracket **9** on the other side of the processing unit. The motor axis of the color image-forming motor **1** penetrates the bracket **9**. Teeth are formed on the circumferential surface of the motor shaft to form a motor gear **2**.

(51) Between the bracket **9** and a side panel **10** on the recess side facing the surface of the bracket **9** on the processing unit side, a photoconductor gear **3Y** for yellow (Y) color, a photoconductor gear **3M** for magenta (M) color, a photoconductor gear **3C** for cyan (C) color, and an idler gear **11** are arranged. The photoconductor gear **3Y** of yellow (Y) color and the photoconductor gear **3M** of magenta (M) color engages with the motor gear **2** of the color image-forming motor **1**. The idler gear **11** according to the present embodiment engages with the photoconductor gear **3M** for magenta (M) color and the photoconductor gear **3C** for cyan (C) color.

(52) The multiple photoconductor gears **3Y**, **3M**, and **3C** are fixed to the rotation axes **17Y**, **17M**, and **17C** that are rotatably supported by the bracket **9** and the side panel **10** on the recess side. To the ends of the multiple rotation axes **17Y**, **17M**, and **17C**, driving photoconductor joints **112Y**, **112M**, and **112C** are attached. The multiple rotation axes **17Y**, **17M**, and **17C** are provided with electromagnetic clutches **7Y**, **7M**, and **7C** and development output gears **6Y**, **6M**, and **6C**.

(53) The color-printing driver device **110** according to the present embodiment is provided with driving development gears **8Y**, **8M**, and **8C**, and the multiple driving development gears **8Y**, **8M**, and **8C** are rotatably supported by the support axes arranged on the side panel **10** on the recess side. In the driving development gears **8Y**, **8M**, and **8C**, a plurality of driving development joints **8b** and a plurality of development gears **8a** that engage with the development output gears **6Y**, **6M**, and **6C** are formed as a single integrated unit.

(54) At the ends of the photoconductors **41Y**, **41M**, and **41C** on the recess side, driven photoconductor joints **124Y**, **124M**, and **124C** that are coupled to the driving photoconductor joints **112Y**, **112M**, and **112C** are arranged. Those driven photoconductor joints **124Y**, **124M**, and **124C** are supported by a plurality of cases **126Y**, **126M**, and **126C** of the image forming unit in a

pivotal manner through a bearing.

(55) At the ends on the recess side, the developer conveying screws **55Y**, **55M**, and **55C** of the developing devices **50Y**, **50M**, and **50C** are provided with driven development gears **180Y**, **180M**, and **180C** are arranged in which a driven gear **180b** and a driven development joint **180a** are formed as a single integrated unit. The developing roller **51Y** and developing rollers **51M** and **51C** are provided with developing roller gears **181Y**, **181M**, and **181C** at their ends on the recess side. The developing roller gears **181Y**, **181M**, and **181C** engage with the driven gears **180b** of the driven development gears **180Y**, **180M**, and **180C**. The driven development joints **180a** are connected to the driving development joints **8b** of the driving development gears **8Y**, **8M**, and **8C**.

(56) As illustrated in FIG. 3, the color photoconductors **41C**, **41M**, and **41Y** and the color developing devices **50Y**, **50M**, and **50C** are all driven to rotate by the color image-forming motor **1** that is a single unit of motor. Due to such a configuration, the number of motors can be reduced, and the manufacturing cost of the toner supply device or the image forming apparatus can be reduced. In regard to black (K) color toner, the photoconductor **41K** and the multiple rotors of the developing device **50K** including the developing roller **51** and the developer conveying screw **55** are driven to rotate by a monochrome image-forming motor.

(57) The toner supply devices are described below in detail. As described above, the toner supply devices according to the present embodiment include the YM color toner supply device **60YM** that supplies the developing device **50Y** that handles yellow (Y) color toner and the toner to the developing device **50M** that handles magenta (M) color toner and the CK color toner supply device **60CK** that supplies the developing device **50C** that handles cyan (C) color toner and the toner to the developing device **50K** that handles black (K) color toner. The YM color toner supply device **60YM** has a configuration or structure similar to that of the CK color toner supply device **60CK**, except for the kind of toner to be supplied. Accordingly, only the CK color toner supply device **60CK** is referred to in the following description of the present disclosure.

(58) FIG. 4 is a schematic diagram of the C color toner supply unit **160C** of the CK color toner supply device **60CK**, according to the present embodiment.

(59) FIG. 5 is a schematic partial sectional view of the C color toner supply unit **160C**, according to the present embodiment.

(60) The K color toner supply unit **160K** has a configuration or structure similar to that of the C color toner supply unit **160C**. The Y color toner supply unit **160Y** and the M color toner supply unit **160M** of the YM color toner supply device **60YM** have a configuration or structure similar to that of the C color toner supply unit **160C**.

(61) The C color toner supply unit **160C** includes, for example, a conveyance nozzle **611C** that serves as a conveyance pipe, a toner conveying screw **614C**, and the downward toner conveyance path **64C**. Once the toner container **32C** moves in the direction of arrow Q as illustrated in FIG. 4 and is mounted on the toner container mount **70** (see FIG. 1) of the printer unit **100**, the conveyance nozzle **611C** of the toner supply device **60C** is inserted from the front end of the toner container **32C** in conjunction with the mounting operation. By so doing, the interior of the toner container **32C** communicates with the conveyance nozzle **611C**.

(62) The toner container **32C** according to the present embodiment is an approximately cylindrical toner bottle. Typically, the toner container **32C** includes a front-end container cover **34C** held in the toner container mount **70** (see FIG. 1) in an irrotational manner, and a container body **33C** integrally molded with a container gear **301C**. The container body **33C** is held to be rotatable relative to the front-end container cover **34C**.

(63) As the container gear **301C** provided for the container body **33C** is driven to rotate by a supply driving unit **91** as will be described later in detail, the container body **33C** is driven to rotate in the direction indicated by arrow A illustrated in FIG. 4. As the container body **33C** itself rotates, by a spiral rib **302C** that is formed on the inner circumference of the container body **33C** to have a helical shape, the toner stored inside the container body **33C** is conveyed in the longitudinal

direction of the container body **33C** from the left to the right as illustrated in FIG. **4**.

(64) On the right side of the container body **33C** near the front-end container cover **34C** as illustrated in FIG. **4**, a pump-up unit **304a** is provided that pumps the toner that is conveyed toward the front-end container cover **34** upward by the rotation of the container body **33C**. The pump-up unit **304a** pumps up the toner above the conveyance nozzle **611C** that is inserted into the toner container **32C**, and the toner drops into a nozzle hole **610** arranged at the end of the conveyance nozzle **611C** near the toner container **32C**. As a result, the toner is supplied to the conveyance nozzle **611C**.

(65) The toner conveying screw **614C** is arranged inside the conveyance nozzle **611C**, and the supply driving unit **91** drives the conveying screw gear **605C** to rotate. As a result, the toner conveying screw **614C** is driven to rotate to convey the toner supplied to the conveyance nozzle **611C**. The downstream end of the conveyance nozzle **611C** in the conveyance direction is connected to the downward toner conveyance path **64C**. The toner that is conveyed by the toner conveying screw **614C** falls off by its own weight along the downward toner conveyance path **64C**, and is supplied to the developing device **50C**.

(66) As illustrated in FIG. **4**, the supply driving unit **91** is controlled by the controller **90**. As the toner conveying screw **614C** that is arranged inside the conveyance nozzle **611C** is driven to rotate, the toner is supplied. For this reason, the controller **90** can calculate the amount of toner supply from the toner container **32C** and the amount of toner supply to the developing device based on the revolutions per minute (rpm) of the toner conveying screw **614C** and the drive time of the toner conveying screw **614C**. As the revolutions per minute (rpm) of the toner conveying screw **614C** is constant in the present embodiment, the controller **90** estimates the amount of toner supply based on the drive time of the toner conveying screw **614C**. More specifically, the controller **90** estimates the amount of the toner that remains in the toner container based on the accumulated drive time, and predicts a toner end in which no toner is left in the toner container. When it is determined that the toner end has come, the controller **90** controls the display unit of the copier **500** to display a message prompting a replacement of the toner container. By driving the toner conveying screw to rotate for a desired driving time, a desired amount of toner is supplied to each developing device.

(67) The supply driving unit **91** of the CK color toner supply device **60CK** is described below. FIG. **6** and FIG. **7** are schematic diagrams each of which illustrates the supply driving unit **91**, according to the present embodiment.

(68) FIG. **6** is a schematic diagram illustrating how driving force is conveyed to the K color toner supply unit **160K**, according to the present embodiment.

(69) FIG. **7** is a schematic diagram illustrating how the driving force is conveyed to the C color toner supply unit **160C**, according to the present embodiment.

(70) Typically, the supply driving unit **91** is provided with a supply motor **603** such as a stepping motor, the K color drive transmission unit **191K**, a cyan (C) color drive transmission unit **191C**, and the switching gear **623**.

(71) The switching gear **623** engages with the motor gear **603a** of the supply motor **603**. The rotation axis **623a** of the switching gear **623** is supported through the bearing **623b** by a long hole **627** that is formed on a board on the image forming apparatus side and extends in the right and left directions as illustrated in FIG. **6** or FIG. **7**. As illustrated in FIG. **6**, when the driving force is conveyed to the K color toner supply unit **160K** for black (K) color toner, the bearing **623b** contacts the left end of the long hole **627** as illustrated in FIG. **6**, and the switching gear **623** engages with the input gear **624K** of the K color drive transmission unit **191K**.

(72) By contrast, as illustrated in FIG. **7**, when the driving force is conveyed to the C color toner supply unit **160C** for cyan (C) color, the bearing **623b** contacts the right end of the long hole **627** as illustrated in FIG. **7**, and the switching gear **623** engages with the input gear **624C** of the C color drive transmission unit **191C**.

(73) Typically, the K color drive transmission unit **191K** includes an input gear **624K**, a bottle drive

gear **626K**, an idler gear **604K**, and a conveying screw gear **605K**. The bottle drive gear **626K** is fixed to the axis of a bottle input gear that engages with the container gear **301** (see FIG. 4), and engages with the input gear **624K** and the idler gear **604K**. The conveying screw gear **605K** engages with the idler gear **604K**.

(74) The C color drive transmission unit **191C** includes the input gear **624C**, a bottle drive gear **626C**, an idler gear **604C**, and the conveying screw gear **605C**. Moreover, the C color drive transmission unit **191C** includes an input idler gear **625C** that engages with the input gear **624C** and the bottle drive gear **626C**. Due to such a configuration, the rotation directions of the container bodies **33K** and **33C** can be made parallel to the rotation directions of the toner conveying screws **614K** and **614C**.

(75) When the toner is supplied to the developing device **50K** that handles black (K) color toner, the controller **90** causes the supply motor **603** to rotate in a counterclockwise direction as illustrated in FIG. 6. Then, force is applied leftward from the motor gear **603a** to the switching gear **623**, and the switching gear **623** moves inside the long hole **627** toward the leftward as illustrated in FIG. 6 or FIG. 7 and engages with the input gear **624K** of the K color drive transmission unit **191K**. As a result, the driving force of the supply motor **603** is conveyed to the K color drive transmission unit **191K** to drive the container body of the toner container **32K** for black (K) color toner and the toner conveying screw **614K** to rotate, and the toner of black (K) color toner is supplied to the developing device **50K** that handles black (K) color toner.

(76) By contrast, when the toner is supplied to the developing device **50C** that handles cyan (C) color, the controller **90** drives the supply motor **603** to rotate in a clockwise direction as illustrated in FIG. 7, which is reverse to the rotation direction when the black (K) color toner is supplied. Then, force is applied rightward from the motor gear **603a** to the switching gear **623**, and the switching gear **623** moves inside the long hole **627** toward the rightward as illustrated in FIG. 6 and engages with the input gear **624C** of the C color drive transmission unit **191C** (see FIG. 7). Due to such a configuration, the driving force of the supply motor **603** is conveyed to the C color drive transmission unit **191C** to drive the container body **33C** of the toner container **32C** for cyan (C) color toner and the toner conveying screw **614C** to rotate, and the toner of cyan (C) color is supplied to the developing device **50C** that handles cyan (C) color toner.

(77) As described above, according to the present embodiment, the pair of toner supply units can be driven by a single supply motor **603**. Due to such a configuration, the manufacturing cost of the toner supply devices or the image forming apparatus can be reduced compared with toner supply devices or image forming apparatuses in which a supply motor is provided for every toner supply unit. In the present embodiment, the switching gear **623** that engages with the motor gear **603a** is configured to be movable in the right and left directions as illustrated in FIG. 6 or FIG. 7 within a prescribed range. Due to such a configuration, just by switching the direction in which the supply motor is driven to rotate, the drive transmission can be switched between the K color drive transmission unit **191K** and the C color drive transmission unit **191C**. Accordingly, compared with a configuration in which, for example, an electromagnetic clutch is arranged to switch the drive transmission, the drive transmission can be switched at low cost.

(78) FIG. 8 is a diagram illustrating the hardware blocks of the copier **500**, according to the present embodiment.

(79) The copier **500** according to the present embodiment is provided with the controller **90**. The controller **90** includes a central processing unit (CPU) **90a**, a read-only memory (ROM) **90b**, a random-access memory (RAM) **90c**, a printer controller **90d**, an image reading controller **90e**, a storage-device controller **90f**, an input controller **90g**, and an output controller **90h**, and the elements of the controller **90** are connected to each other through a bus.

(80) The printer controller **90d** of the controller **90** serves as an interface for the CPU **90a** to control the multiple hardware components of the printer unit **100** such as the multiple image forming units **46**, the intermediate transfer unit **85**, the exposure device **47**, the fixing device **86**, and the toner

supply devices **60**. The image reading controller **90e** serves as an interface for the CPU **90a** to control the scanner unit **400b** and the document conveyance unit **400a** of the image reading unit **400**.

(81) The storage-device controller **90f** serves as an interface for the CPU **90a** to control a storage device (memory) **93** provided for the copier **500**. The input controller **90g** serves as an interface for the CPU **90a** to control the input device **94**. The output controller **90h** serves as an interface for the CPU **90a** to control the output device **95**.

(82) The controller **90** that includes the CPU **90a** controls the entirety of the copier **500**. The CPU **90a** starts the operating system (OS) using the boot program stored in the ROM **90b**. On the OS, the CPU **90a** executes the control program stored in the storage device **93** and the ROM **90b**.

(83) The RAM **90c** is used as a temporary storage area such as a main memory or a work area of the CPU **90a**. The storage device **93** is a readable/writable nonvolatile storage device such as a hard disk drive (HDD). In the storage device **93**, various kinds of data such as programs or various types of application programs used to control the entirety of the copier **500**, data used to manage consumable items, and moving images presenting a series of work required to solve maintenance events are stored. The CPU **90a** accesses the storage device **93** using the storage-device controller **90f**.

(84) The CPU **90a** reads a control program or an application program from the storage device **93** or the ROM **90b**, and executes the program developed in the RAM **90c** to control the copier **500**.

(85) In the copier **500** according to the present embodiment, one CPU uses a program developed in one memory such as a RAM to execute the multiple processes in the flowcharts as will be described later in detail. However, no limitation is indicated thereby, and other kinds of modes may be adopted. For example, a plurality of processors, RAMs, ROMs, and storage devices may cooperate to execute the multiple processes in the flowcharts as will be described later in detail. Alternatively, some of the processes may be executed using hardware circuitry such as an application-specific integrated circuit (ASIC) or a field-programmable gate array (FPGA).

(86) The CPU **90a** according to the present embodiment controls the scanner unit **400b** through the image reading controller **90e** to scan images on the sheet of document and generate image data. The CPU **90a** according to the present embodiment cooperates with the multiple units of hardware of the printer controller **90d** and the printer unit **100** to form an image on a recording medium such as a sheet of paper.

(87) The input controller **90g** connects the input device **94** and the controller **90** to accept instructions given through the input device **94** such as a touch panel and hard keys. The output controller **90h** connects the output device **95** and the controller **90**, and controls the output device **95** provided with a display unit such as a liquid crystal display (LCD) and a cathode-ray tube (CRT) display to display an operation screen or moving images to a user. In the present embodiment, the output device **95** is described as a display unit that produces a visual output. However, no limitation is indicated thereby, and in addition to or in place of the visual output, the output device **95** may be provided with a loudspeaker that produces a voice output or audio output. In addition to or in place of, for example, a touch panel and hard keys, the input device **94** may be provided with a microphone through which a voice input is made.

(88) As illustrated in FIG. 3, in the copier **500** according to the present embodiment, the photoconductors **41** and the rotors of the developing devices **50** (including the developing rollers **51** and the developer conveying screws **55**) are driven by a single image-forming motor for the purposes of reducing the manufacturing cost of the toner supply devices or the image forming apparatus. When the toner is supplied to each of the developing devices **50**, the developing devices **50** need to be driven to stir and convey the toner that is supplied to each of the developing devices **50** using the developer conveying screw **55**. By so doing, the toner concentration in the developing devices **50** is made even. Under these circumstances, in the image forming apparatus according to the present embodiment where the photoconductors **41** and the rotors of the developing devices **50**

including the developing roller **51** and the developer conveying screw **55** are driven by a single motor, the photoconductors **41** are also driven to rotate during the toner supply. For this reason, if the toner is supplied while an image is not being formed, the consumption of the photoconductors **41** speeds up, and the useful life of the photoconductors **41** is shortened. In order to handle such a situation, in the present embodiment, toner is supplied to each of the developing devices **50** during the image-forming operation.

(89) FIG. **9** is a flowchart of the processes of setting the drive time of a toner supply unit of each color or the drive time of the toner conveying screw **614** during the toner supply, according to the present embodiment.

(90) Once the image-forming operation is completed, firstly, in step **S1**, the controller **90** (see FIG. **4**) calculates the amount of the toner consumed during the current image-forming operation based on the image data used by the exposure device **47**. There are some cases in which toner adheres to areas other than the areas for latent images on the photoconductors **41**. For example, toner may adhere to areas outside the image-forming area on both sides of the photoconductors **41** in the axial direction as scumming. In order to handle such a situation, the controller **90** calculates the amount of the toner that adheres to areas outside the image-forming area as scumming based on the travel distance of the photoconductors **41** in the current image-forming operation, and adds the calculated amount of toner to amount of toner consumption.

(91) Subsequently, in step **S2**, the controller **90** calculates the length of time during which toner conveying screws **614** are driven, based on the calculated amount of toner consumption. As described above, toner is supplied by driving the toner conveying screws **614** arranged inside the conveyance nozzles **611** to rotate. For this reason, by controlling the revolutions per minute (rpm) of the toner conveying screws **614** and the drive time of the toner conveying screws **614**, a desired amount of toner can be supplied to the developing devices **50**. As described above, the revolutions per minute (rpm) of the toner conveying screws **614** is constant in the present embodiment. In view of these circumstances, the drive time of the toner conveying screw **614** is set based on the calculated amount of consumed toner. By so doing, the toner of the amount of toner consumption can be supplied to each one of the developing devices **50**.

(92) Subsequently, in step **S3**, the controller **90** checks whether the toner supply is completed during the current image-forming operation. As described above, the toner supply device according to the present embodiment is configured to switch the rotation direction of the supply motor **603** to switch the toner supply unit to be driven. Due to such a configuration, in the CK color toner supply device **60CK**, the K color toner supply unit **160K** and the C color toner supply unit **160C** are sequentially driven. For this reason, the length of time required to supply a desired amount of toner to each of the developing device **50K** that handles black (K) color toner and the developing device **50C** that handles cyan (C) color toner is defined as follows. In other words, the length of time required to supply a desired amount of toner is the total time of the drive time of the toner conveying screw **614K** for black (K) color toner, which is set based on the amount of toner consumption, and the drive time of the toner conveying screw **614C** for cyan (C) color toner, which is set based on the amount of toner consumption.

(93) The toner supply available time that corresponds to the length of time of the image-forming operation varies depending on the number of sheets to be printed, the conveyance direction of the sheet P, or the length of the sheet P. For this reason, there are some cases in which the length of time of the image-forming operation becomes shorter than the above length of time required to supply toner. In such cases, the toner supply control is terminated when the image-forming operation is terminated, and there is some concern that a desired amount of toner cannot be supplied to the developing device.

(94) In order to handle such a situation, when the toner supply is not completed during the current image-forming operation (“NO” in step **S3**), in step **S4**, the controller **90** adds the lacking drive time to the set drive time based on the amount of toner consumption. The lacking drive time is a

value of the difference between the target drive time of the toner conveying screws **614** in the current toner supply control and the actual drive time of the toner conveying screws **614** in the current image-forming operation. The drive time to which the lacking drive time is added is stored in a nonvolatile memory as the drive time of the toner conveying screw, which corresponds to the amount of toner supply, in the toner supply control of the next image-forming operation.

(95) By contrast, when the toner supply is completed during the current image-forming operation (“NO” in step **S3**), the drive time set in step **S2** is determined to be the drive time of the toner conveying screw in the toner supply control of the next image-forming operation, and that drive time is stored in a nonvolatile memory.

(96) Subsequently, in step **S5**, the controller **90** checks whether the drive time of the toner conveying screw in the toner supply control of the next image-forming operation exceeds the threshold. When the drive time exceeds the threshold, the degree of toner concentration in the developing device **50** may have decreased to a level at which a desired level of image density cannot be obtained. In order to handle such a situation, when the drive time exceeds the threshold (“YES” in step **S5**), in step **S6**, the controller **90** performs forced toner supply control, for example, before the next image-forming operation starts. In the forced toner supply control, the toner conveying screws are driven to rotate for a driving time equal to or exceeding the threshold. Due to such a configuration, image formation can be performed after the degree of toner concentration of the developer in the developing device **50** is recovered, and an image with a desired level of image density can be obtained.

(97) For example, the developing device may be provided with a toner concentration sensor to detect the degree of toner concentration of the developer **G**, and the forced toner supply control may be performed when the degree of toner concentration of the toner concentration sensor becomes equal to or less than the threshold. In such a configuration, image patterns are at prescribed time intervals, and the amount of adhered toner in the image patterns is detected by the toner adhesion-amount sensor **310**. When the amount of adhered toner in the image patterns falls below the threshold and the image density does not recover by adjusting the image-forming condition such as a developing bias, the forced toner supply control may be performed.

(98) As the forced toner supply control is performed, the photoconductors **41** rotate in operations other than the image-forming operation. For this reason, when the forced toner supply control is performed frequently, the useful life of the photoconductors **41** is shortened. While the CK color toner supply device **60CK** performs the toner supply in which the toner supply of black (K) color toner and the toner supply of cyan (C) color toner are sequentially performed only once during the image-forming operation and the drive time of the toner conveying screws for multicolor toner, which corresponds to the amount of toner supply is satisfied, there is some concern that the forced toner supply control is performed frequently. For this reason, it is desired that the length of time of the toner supply by the CK color toner supply device **60CK** be shortened and the toner supply be performed a plurality of times. By so doing, the drive time of the toner conveying screws for multicolor toner, which corresponds to the amount of toner supply is satisfied.

(99) FIG. **10A** is a timing chart of cases where the toner supply in which the toner supply of black (K) color toner and the toner supply of cyan (C) color toner are sequentially performed is performed only once, according to the present embodiment.

(100) FIG. **10B** is a timing chart of cases in which the toner supply is performed a plurality of times, according to the present embodiment.

(101) In the present embodiment described with reference to FIG. **10A** and FIG. **10B**, the length of time of the image-forming operation is shorter by 3 seconds (S) than the length of time required to supply a desired amount of toner to each of the developing device **50K** that handles black (K) color toner and the developing device **50C** that handles cyan (C) color toner.

(102) In the toner supply control described above with reference to FIG. **10A**, firstly, the K color toner supply unit **160K** drives the toner conveying screw **614K** to rotate for the drive time stored in

the nonvolatile memory to supply a desired amount of toner to the developing device **50K** that handles black (K) color toner. Secondly, in the toner supply control described above with reference to FIG. **10A**, the rotation direction of the supply motor **603** is switched, and the C color toner supply unit **160C** drives the toner conveying screw **614C** to rotate for the drive time stored in the nonvolatile memory. As a result, a desired amount of toner is supplied to the developing device **50C** that handles cyan (C) color toner.

(103) As illustrated in FIG. **10A**, the toner supply available time or the length of time of the image-forming operation is substantially equal to the drive time or the supplying time to supply toner of the toner conveying screw **614K** for black (K) color toner. For this reason, toner is not at all supplied to the developing device **50C** that handles cyan (C) color. As a result, for cyan (C) color toner, the amount of toner supply for the next time to which the amount of lacking toner, which corresponds to, for example, the lacking drive time of 3 seconds, is added, which corresponds to the drive time of the toner conveying screw **614C**, exceeds the threshold (“YES” in step **S5** of FIG. **9**), and the forced toner supply control is performed before the next image-forming operation.

(104) In the toner supply control described above with reference to FIG. **10B**, the length of time of one-time toner supply is shortened, and the toner supply is performed a plurality of times. By so doing, a desired amount of toner is supplied to each developing device. More specifically, in the toner supply according to the present embodiment, the maximum supplying time is determined in advance and the toner supply of black (K) color toner and the toner supply of cyan (C) color toner are switched when the supplying time to supply toner exceeds the maximum supplying time. In the toner supply control according to the present embodiment, such toner supply of black (K) color toner and cyan (C) color toner is performed repeatedly to satisfy the drive time stored in the nonvolatile memory. Accordingly, a desired amount of toner is supplied to each one of the developing devices **50K** and **50C**.

(105) As described above, by performing the toner supply of black (K) color toner and cyan (C) color toner a plurality of times to satisfy the drive time stored in the nonvolatile memory, the toner of both black (K) color and cyan (C) color can be supplied to the developing devices **50**, which is different from the toner supply control described above with reference to FIG. **10A**.

(106) In the toner supply control described above with reference to FIG. **10B**, the lacking drive time can be divided into the lacking drive time for black (K) color toner and the drive time for cyan (C) color toner. More specifically, in FIG. **10B**, the lacking drive time for black (K) color toner is 2 seconds (S), and the lacking drive time for cyan (C) color toner is 1 second (S). As described above, as the lacking drive time is divided into the lacking drive time for black (K) color toner and the drive time of cyan (C) color toner, the amount of toner supply of both cyan (C) color toner and black (K) color toner for the next time to which the amount of lacking toner, which corresponds to the lacking drive time, is added can be prevented from exceeding the threshold. Note also that the amount of toner supply for the next time corresponds to the drive time of the toner conveying screw. Due to such prevention from exceeding the threshold in regard to both cyan (C) color toner and black (K) color toner, the forced toner supply control can be prevented from being performed before the next image-forming operation. As a result, the photoconductors **41** are prevented from being driven except during the image-forming operation, and a reduction in the useful life of the photoconductors can be prevented.

(107) However, as illustrated in FIG. **10B**, there is room for improvement in the toner supply device that performs the toner supply control in which the length of time of one-time toner supply is shortened and the toner supply is performed a plurality of times to supply a desired amount of black (K) color toner and cyan (C) color toner. More specifically, as illustrated in FIG. **11**, when the length of time of the image-forming operation or the toner supply available time is very short and the length of time of the image-forming operation is equal to or shorter than the maximum supplying time, there are some cases in which toner is not at all supplied to the developing device **50C** that handles cyan (C) color. In FIG. **11**, the length of time of the image-forming operation is

equal to the maximum supplying time. The length of time of the image-forming operation is the length of time elapsed since the photoconductors **41** started rotating for image formation by the time at which the photoconductors **41** stops rotating. In other words, the length of time of the image-forming operation is the drive time of the image-forming motor.

(108) When the length of time of the image-forming operation is very short, the amount of toner consumption is small. Accordingly, if the image-forming operation whose operation time is very short is performed only once, the amount of toner in the developing device **50** that handles cyan (C) color toner does not decrease to such an extent that the image quality is affected. In other words, the amount of toner supply for the next time to which the amount of lacking toner is added or the drive time of the toner conveying screw to which the lacking drive time is added does not exceed the threshold. However, as illustrated in FIG. **11**, when the image-forming operation whose operation time is very short continues several times, the amount of lacking toner that corresponds to the lacking drive time is accumulated. As a result, the amount of toner supply for the next time, which corresponds to the drive time of the toner conveying screw, exceeds the threshold, and the amount of toner in the developing device **50** decreases to such an extent that the image quality is affected. Accordingly, for cyan (C) color toner, the forced toner supply control is performed in an undesired manner.

(109) In order to handle such a situation, the maximum supplying time in the toner supply may be made equal to or shorter than the minimum time of image-forming operation, and the length of time of the toner supply may be made shorter than the minimum time of the image-forming operation. The minimum time of the image-forming operation according to the present embodiment indicates the length of time when an image is formed on a single sheet P whose length in the conveyance direction is minimum necessary to be conveyed in the printer unit **100** according to the present embodiment.

(110) However, if the maximum supplying time is shortened, due to the rotation of the container body **33** of the toner container **32**, the container body **33** stops rotating and the operation is switched to the toner supply of the next color before the toner inside the toner container **32** is sufficiently loosened. In other words, the toner conveying screw tends to stop rotating before the toner in the conveyance nozzles **611** is sufficiently loosened by the toner conveying screws **614**. As a result, the toner in the toner container or the conveyance nozzle is not sufficiently loosened over time, and the toner in the toner container and the conveyance nozzle tends to be tightened. As a result, the toner cannot easily be conveyed, and the amount of toner supply per unit time tends to be reduced.

(111) The length of time of the image-forming operation or the toner supply available time may be calculated based on, for example, the conveyance direction of the sheet on which images are to be formed, the length of the sheet, or the number of sheets to be printed continuously, and the maximum supplying time may be determined based on the obtained length of time of the image-forming operation. However, in such a configuration, the operation of toner supply control tends to become complicated.

(112) In order to handle such a situation, in the present embodiment, the maximum supplying time in the toner supply for the first time is made shorter than the maximum supplying time in the toner supply for the second and following times.

(113) FIG. **12** is a timing chart of toner supply control according to the present embodiment.

(114) In the present embodiment described with reference to FIG. **12**, the maximum supplying time is reduced to half the minimum time of the image-forming operation or shorter, and the length of time of the toner supply for the first time after the toner supply control starts is made shorter than the minimum time of the image-forming operation.

(115) By making the length of time of the toner supply for the first time shorter than the minimum time of the image-forming operation, even when the length of time of the image-forming operation is the shortest as illustrated in FIG. **12**, the toner of both cyan (C) color and black (K) color can be

supplied to the developing devices. Due to such a configuration, the forced toner supply control that is performed when the image-forming operation where the length of time of the image-forming operation is very short continues can be prevented. More specifically, in the case of FIG. 11, only the amount of lacking toner of cyan (C) color, which corresponds to the lacking drive time, is accumulated. By contrast, in the present embodiment, the amount of lacking toner, which corresponds to the lacking drive time, that is accumulated is distributed into that of cyan (C) color and that of black (K) color.

(116) In the present embodiment described with reference to FIG. 11, for example, when the shortest image-forming operation is performed continuously three times, the amount of toner supply of cyan (C) color toner for the next time, which corresponds to the drive time of the toner conveying screw, exceeds the threshold, and the forced toner supply control is performed. By contrast, in the present embodiment, the amount of lacking toner, which corresponds to the lacking drive time, that is accumulated is distributed into that of cyan (C) color and that of black (K) color. Due to such a configuration, even when the shortest image-forming operation is performed continuously three times, the amounts of toner supply of both cyan (C) color and black (K) color for the next time, which corresponds to the drive time of the toner conveying screw, do not exceed the threshold. Accordingly, compared with the present embodiment described with reference to FIG. 11, the number of occurrences in which the forced toner supply control is performed can be reduced and a reduction in the useful life of the photoconductors can be prevented in the present embodiment.

(117) The maximum supplying time in the toner supply for the second and following times is made longer than the maximum supplying time in the toner supply for the first time. Due to such a configuration, the length of time of the image-forming operation, which corresponds to the toner supply available time, is sufficiently long. Moreover, when the toner supply is performed a plurality of times, the toner in the toner container and the conveyance nozzle can be loosened as desired in the toner supply for the second and following times. Due to such a configuration, the toner in the toner container and the conveyance nozzle can be prevented from being tightened over time, and a reduction in the amount of toner supply per unit time can be prevented. Moreover, as the toner supply control can be performed evenly regardless of the length of time of the image-forming operation, the toner supply control can be simplified, and the toner supply device and the image forming apparatus according to the above embodiments of the present application can be produced at low cost.

(118) FIG. 13 is a flowchart of the processes of toner supply control, according to the present embodiment.

(119) FIG. 14 is a flowchart of toner supply according to the present embodiment.

(120) As illustrated in FIG. 13, when the driving of the image-forming motor and the rotation of the photoconductors 41 start and the image-forming operation starts, toner supply control starts. When the toner supply control starts, in step S11, the controller 90 defines the maximum supplying time in the toner supply for the first time, which is stored in the nonvolatile memory of the controller 90, as the maximum supplying time. Then, in step S12, the controller 90 performs the toner supply illustrated in FIG. 13. When the toner supply of black (K) color toner and cyan (C) color toner is not completed in the toner supply for the first time (“NO” in step S13), in step S14, the controller 90 defines the maximum supplying time in the toner supply for the second and following times, which is stored in the nonvolatile memory of the controller 90, as the maximum supplying time. In step S15, the controller 90 performs the toner supply illustrated in FIG. 13. Until the toner supply of both black (K) color toner and cyan (C) color toner is completed (“YES” in step S13), the controller 90 repeats the processes in steps S14 and S15.

(121) As illustrated in FIG. 14, in the toner supply according to the present embodiment, when the toner supply of black (K) color toner is completed (“YES” in step S21), in steps S27 to S30, the toner supply of cyan (C) color toner is performed without driving the K color toner supply unit

160K.

(122) By contrast, when the toner supply of black (K) color toner is not completed (“NO” in step S21), in step S22, the K color toner supply unit **160K** is driven to supply the K color toner. When the total drive time of the toner conveying screw, which corresponds to the supplying time to supply toner, satisfies the set drive time before exceeding the set maximum supplying time in the processes described with reference to FIG. 9 (“YES” in step S23), it is determined that a desired amount of black (K) color toner is supplied and the K color toner supply is completed (“YES” in step S25). When the toner supply of cyan (C) color toner is completed (“YES” in step S26), the toner supply is completed. In such cases, the toner supply of both black (K) color toner and cyan (C) color toner is completed, and the toner supply control is completed (“YES” in step S13 of FIG. 13). By contrast, when the toner supply of cyan (C) color toner is not completed (“NO” in step S26), in steps S27 to S30, the toner supply of cyan (C) color toner is performed.

(123) When the supplying time to supply black (K) color toner, which corresponds to the drive time of the toner conveying screw **614K**, exceeds the maximum supplying time (“NO” in step S23 and “YES” in step S24) and the toner supply of cyan (C) color toner is completed (“YES” in step S26), the toner supply is completed. By contrast, when the toner supply of cyan (C) color toner is not completed, in steps S27 to S30, the toner supply of cyan (C) color toner is performed. In these cases, a desired amount of black (K) color toner is not supplied to the developing device, and the toner supply is performed a plurality of times (“NO” in step S13 of FIG. 13).

(124) In a similar manner to the case of black (K) color toner, in the toner supply of cyan (C) color toner, when the total drive time of the toner conveying screw, which corresponds to the supplying time to supply toner, satisfies the set drive time before exceeding the set maximum supplying time in the processes described with reference to FIG. 9 (“YES” in step S28), in step S30, it is determined that a desired amount of cyan (C) color toner is supplied and the C color toner supply is completed. When the toner supply of black (K) color toner is completed in the above steps, it is determined that the toner supply control is completed (“YES” in step S13 of FIG. 13). By contrast, when the toner supply of black (K) color toner is not completed, the toner supply for the next time will be only black (K) color toner.

(125) When the maximum supplying time for the cyan (C) color toner has passed (“YES” in step S29), the toner supply is terminated without completing the toner supply. As the toner supply of at least cyan (C) color toner is not completed at this point in time (“NO” in step S13 of FIG. 13), in steps S14 and S15, the toner supply continues.

(126) In the above-described embodiments of the present disclosure, the maximum supplying time for the second and following times is even. However, no limitation is indicated thereby, and the maximum supplying time may be made longer step by step. By so doing, chances of the developing devices supplying no toner in the toner supply for the second time can be reduced, and the toner inside the toner container or the conveyance nozzle can be well loosened compared with the toner supply for the first time.

(127) The ratio of the maximum supplying time to supply the black (K) color toner to the maximum supplying time to supply the cyan (C) color toner in the toner supply for the first time may be changed based on the drive time of the toner conveying screws for multicolor toner, which corresponds to the amount of required toner supply. For example, when the calculated drive time of the toner conveying screw for the next time for cyan (C) color toner has a sufficient margin for the threshold and the calculated drive time of the toner conveying screw for the next time for black (K) color toner has a little margin for the threshold, the ratio of the maximum supplying time to supply the black (K) color toner in the toner supply for the first time is increased. Due to such a configuration, when the image-forming operation where the length of time of the image-forming operation is very short continues, the number of occurrences in which the forced toner supply control is performed can further be reduced.

(128) The above-described embodiments of the present disclosure are illustrative and do not limit

the present disclosure. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present disclosure.

(129) The embodiments described above are given as an example, and unique advantageous effects are achieved for each of the following modes given below.

(130) First Aspect

(131) The toner supply device **60** includes a plurality of toner containers such as the multiple toner containers **32C** and **32K** that contain toner used to form a latent image on a latent-image bearer such as the photoconductor **41**, and perform toner supply control in which toner is supplied from a plurality of toner containers to a plurality of developing devices. In the toner supply control, toner supply is repeatedly performed in which one of the plurality of toner containers supplying the toner is switched to another one of the plurality of toner containers when a supplying time to supply the toner from one of the plurality of toner containers to corresponding one of the plurality of developing devices exceeds a prescribed maximum supplying time, and the prescribed maximum supplying time from one of the plurality of toner containers, in the toner supply performed for a first time after the toner supply control starts, is shorter than the prescribed maximum supplying time from one of the plurality of toner containers in the toner supply performed for second and following times.

(132) When the toner supply control is performed during the image-forming operation, the length of time of the image-forming operation varies depending on the conveyance direction of the sheet (P), the length of the sheet (P), or the number of sheets to be printed continuously, and the toner supply available time also varies accordingly. For this reason, there are some cases in which the length of time of the image-forming operation becomes shorter than the length of supplying time to supply a desired amount of toner to each developing device, which corresponds to the length of time of the toner supply control. In such cases, the toner supply control is terminated when the image-forming operation is completed. In cases where the toner supply control is performed as follows, there may be a developing device to which toner is not at all supplied when the length of time of the image-forming operation is very short. In such cases, for example, toner supply is repeatedly performed in which one of the multiple toner containers that supply the toner is switched to another one of the multiple toner containers when the supplying time to supply toner from one of the multiple toner containers to corresponding one of the multiple developing devices exceeds the prescribed maximum supplying time (see FIG. **10B**). If such toner supply control is performed, the length of time of the image-forming is very short, and once the image-forming operation is completed before the toner supply performed for the first time after the toner supply control starts ends, there may be a developing device to which toner is not at all supplied (see, for example, FIG. **11**). If such an image-forming operation where the length of time of the image-forming operation is very short is continuously performed, there may be a developing device that lacks toner, and the image quality may deteriorate as, for example, an image may be blurred and a desired level of image density is not achieved.

(133) In order to handle such a situation, in the toner supply device **60** according to the first aspect of the present disclosure, as illustrated in FIG. **12**, the prescribed maximum supplying time to the toner container in the toner supply performed for the first time after the toner supply control starts is made shorter than the maximum supplying time in the toner supply for the second and following times. Due to such a configuration, in the toner supply for the first time, the toner can be supplied in a short period of time to all the developing devices to which the toner is to be supplied.

Accordingly, cases in which toner is not at all supplied to the developing device when the length of time of the image-forming operation is very short can be prevented, and cases in which the developing device lacks toner can be prevented.

(134) By making the prescribed maximum supplying time for the toner container in the toner

supply for the second and following times longer than the prescribed maximum supplying time from one of the plurality of toner containers in the toner supply for the first time, the length of time of the image-forming operation can be made relatively long. Moreover, when the toner supply is performed for the second time, the toner inside the multiple toner containers and the toner inside the toner supply path can be stirred and loosened as desired. Accordingly, the toner inside the multiple toner containers and the toner inside the toner supply path can be prevented from being tightened over time. As a result, the toner inside the multiple toner containers and the toner inside the toner supply path can be prevented from being tightened upon conveyance, and a reduction in the amount of toner supply per unit time can be prevented.

(135) Second Aspect

(136) In the toner supply device **60** according to the first aspect of the present disclosure, the toner supply control is performed during an image-forming operation in which an image is formed on a sheet using the toner, and a time of the toner supply performed for the first time is shorter than a minimum time of the image-forming operation.

(137) Due to such a configuration, even in the shortest image-forming operation, the toner can be supplied to each one of the developing devices, and the amount of toner in the multiple developing devices **50** can be prevented from decreasing to such an extent that the image quality is affected.

(138) Third Aspect

(139) In the toner supply device **60** according to the first or second aspect of the present disclosure, the toner supply control is performed during an image-forming operation in which an image is formed on a sheet using the toner, and when an amount of toner in one of the plurality of developing devices is reduced to a level affecting the image, the circuitry is configured to perform forced toner supply control to supply the toner to one of the plurality of developing devices whose amount of toner is reduced to the level affecting the image, except during the image-forming operation.

(140) Due to such a configuration, a reduction in the image density of the image formed on the sheet due to an insufficient amount of toner in the developing devices can be prevented.

(141) Fourth Aspect

(142) In the toner supply device **60** according to any one of the first to third aspects of the present disclosure, the toner in the pair of toner containers is supplied to corresponding ones of the plurality of developing devices using a single motor such as the supply motor **603**, and the motor is controlled in the toner supply for two of the plurality of toner containers to supply the toner in one of the plurality of toner containers to corresponding one of the plurality of developing devices step by step.

(143) Due to such a configuration, as described above in the embodiments of the present disclosure, compared with the toner supply devices or image forming apparatuses in which a motor such as the supply motor **603** is provided for each one of the toner containers, the manufacturing cost of the toner supply devices or the image forming apparatus can be reduced.

(144) Fifth Aspect

(145) In the toner supply device **60** according to the fourth aspect of the present disclosure, one of the plurality of toner containers that supplies the toner is switched by switching a rotation direction of a motor such as the supply motor **603**.

(146) Due to such a configuration, as described above in the embodiments of the present disclosure, a plurality of toner containers that supply toner can sequentially be switched without using an electromagnetic clutch or the like, and a plurality of toner containers that supply toner can be switched at low cost.

(147) Sixth Aspect

(148) An image forming apparatus includes a plurality of image forming units each including a latent-image bearer such as the photoconductor **41** and the developing device **50** to develop a latent image on the latent-image bearer with toner, and a toner supply device to supply the toner to the

developing device of each one of the plurality of image forming units. In such an image forming apparatus, the toner supply device according to any one of the first to sixth aspects is used as the toner supply device.

(149) Due to such a configuration, a reduction in the useful life of latent-image bearers such as the photoconductors can be prevented.

(150) Seventh Aspect

(151) In the image forming apparatus according to the sixth aspect of the present disclosure, the developing device **50** and a latent-image bearer such as the photoconductor **41** of the image forming unit are driven by a single motor.

(152) Due to such a configuration, as described above in the embodiments of the present disclosure, compared with a configuration in which the developing device is driven by a motor different from the motor that drives a latent-image bearer such as the photoconductor **41**, the number of motors can be reduced, and the toner supply device and the image forming apparatus according to the above embodiments of the present application can be produced at low cost.

(153) Any one of the above-described operations may be performed in various other ways, for example, in an order different from the one described above.

(154) Each of the functions of the described embodiments may be implemented by one or more processing circuits or circuitry. Processing circuitry includes a programmed processor, as a processor includes circuitry. A processing circuit also includes devices such as an application-specific integrated circuit (ASIC), digital signal processor (DSP), field-programmable gate array (FPGA), and conventional circuit components arranged to perform the recited functions.

Claims

1. A toner supply device comprising: a plurality of toner containers each containing toner used to form a latent image on a latent-image bearer; and circuitry configured to perform toner supply control to supply toner from the plurality of toner containers to a plurality of developing devices, wherein the toner supply control includes repeating toner supply in which one of the plurality of toner containers supplying the toner is switched to another one of the plurality of toner containers when a supplying time to supply the toner from one of the plurality of toner containers to corresponding one of the plurality of developing devices exceeds a prescribed maximum supplying time, and the prescribed maximum supplying time, in the toner supply performed for a first time after the toner supply control starts, being shorter than the prescribed maximum supplying time in the toner supply performed for a second time.
2. The toner supply device according to claim 1, wherein the prescribed maximum supplying time, in the toner supply performed for the first time after the toner supply control starts, is shorter than the prescribed maximum supplying time in the toner supply performed for the second time and following times.
3. The toner supply device according to claim 1, wherein the circuitry is configured to perform the toner supply control during an image-forming operation in which an image is formed on a sheet using the toner, and wherein a time of the toner supply performed for the first time is shorter than a minimum time of the image-forming operation.
4. The toner supply device according to claim 1, wherein the circuitry is configured to perform the toner supply control during an image-forming operation in which an image is formed on a sheet using the toner, and wherein, when an amount of toner in one of the plurality of developing devices is reduced to a level affecting the image, the circuitry is configured to perform forced toner supply control to supply the toner to one of the plurality of developing devices whose amount of toner is reduced to the level affecting the image, except during the image-forming operation.
5. The toner supply device according to claim 1, wherein the toner in two of the plurality of toner containers is supplied to corresponding ones of the plurality of developing devices using a single

motor, and wherein the circuitry is configured to control the single motor in the toner supply for two of the plurality of toner containers to supply the toner in one of the plurality of toner containers to corresponding one of the plurality of developing devices step by step.

6. The toner supply device according to claim 5, wherein the circuitry is configured to switch a rotation direction of the single motor to switch one of the plurality of toner containers that supplies the toner.

7. An image forming apparatus comprising: a plurality of image forming units each including a latent-image bearer and a developing device to develop a latent image on the latent-image bearer with toner; and a toner supply device including: a plurality of toner containers each containing toner used to form a latent image on a latent-image bearer, and circuitry configured to perform toner supply control to supply toner from the plurality of toner containers to a plurality of developing devices, wherein the toner supply control includes repeating toner supply in which one of the plurality of toner containers supplying the toner is switched to another one of the plurality of toner containers when a supplying time to supply the toner from one of the plurality of toner containers to corresponding one of the plurality of developing devices exceeds a prescribed maximum supplying time, the prescribed maximum supplying time, in the toner supply performed for a first time after the toner supply control starts, being shorter than the prescribed maximum supplying time in the toner supply performed for a second time, the toner supply device supplies the toner to the developing device of each one of the plurality of image forming units.

8. The image forming apparatus according to claim 7, wherein the prescribed maximum supplying time, in the toner supply performed for the first time after the toner supply control starts, is shorter than the prescribed maximum supplying time in the toner supply performed for the second time and following times.

9. The image forming apparatus according to claim 7, wherein the developing device and the latent-image bearer of the image forming unit are driven by a single motor.
